CSE 441: Internet of Things
Week 9 / Connecting Smart Objects

Complete 10-Mark Exam Answers

ZigBee, IEEE 802.15.4, Wi-Fi HaLow, 6LoWPAN, LOADng and RPL

Course	CSE 441 - Internet of Things
Answer Style	Numbered points + explanation + example
Marks	Each question prepared for 10 marks
Source Basis	W9_compressed lecture slides and course outline

Question 1: Explain ZigBee architecture, device roles, routing, and network topologies.

Answer (10 Marks)

Introduction: ZigBee is a low-power and low-data-rate wireless technology built on IEEE 802.15.4. It is suitable for constrained IoT devices that require reliable communication with low energy consumption.

1. **Layered architecture:** The PHY and MAC layers are provided by IEEE 802.15.4, while ZigBee adds the network, security, application support, device object, and application profile functions. Example: A smart-light sensor uses IEEE 802.15.4 radio communication and ZigBee application services.
2. **Application/Profiles layer:** This top layer contains ZigBee or vendor-specific profiles that define device behavior and interoperability. Example: A home-automation profile defines how a light switch communicates with a lamp.
3. **Application Support Sub-layer (APS):** APS interfaces application objects with network services and supports binding and service control. Example: A temperature sensor binds its report to a thermostat application.
4. **ZigBee Device Object (ZDO):** ZDO manages devices, security, policies, discovery, and network participation. Example: A new sensor uses ZDO functions to discover and join the network.
5. **Network and Security layer:** This layer provides network formation, addressing, routing, and security. Example: A router forwards a packet from a sensor to the coordinator.
6. **Device roles:** A ZigBee network includes one coordinator, routers, and end devices. The coordinator forms the network, routers forward packets, and end devices mainly sense or act. Example: In smart agriculture, the gateway is coordinator, relay nodes are routers, and soil sensors are end devices.
7. **AODV routing:** The ZigBee network layer uses Ad Hoc On-Demand Distance Vector routing to discover routes when needed. Example: A route is created only when a sensor must send data through multiple routers.
8. **Star topology:** All end devices communicate through the coordinator; it is simple but coordinator failure affects the network. Example: A small room with one coordinator and several smart bulbs.
9. **Tree and mesh topologies:** Cluster-tree provides hierarchical communication, while mesh offers multiple paths and improved fault tolerance. Example: If one mesh router fails, data may use another path.
10. **Low-power suitability:** Sleeping end devices and short packets reduce energy consumption. Example: A battery-powered door sensor remains asleep until the door opens.

Others / Additional Relevant Points: ZigBee supports scalable IoT networks, but topology selection should depend on range, reliability, cost, and energy requirements.

Conclusion: ZigBee combines IEEE 802.15.4 with network, security, and application services to support star, tree, and mesh IoT networks.

Very Important Exam Suggestion: Draw the ZigBee protocol stack and a small topology diagram. Underline Coordinator, Router, End Device, AODV, APS, and ZDO.

Question 2: Describe the ZigBee IP protocol stack and its physical layer.

Answer (10 Marks)

Introduction: ZigBee IP integrates IPv6-based Internet communication with low-power IEEE 802.15.4 networks. It enables constrained devices to become part of an IP-based IoT system.

1. **ZigBee IP profile:** The top profile, such as Smart Energy 2.0, defines application behavior. Example: A smart meter follows the Smart Energy profile.

2. **Transport layer:** ZigBee IP supports UDP and TCP according to application needs. Example: UDP may carry periodic sensor readings, while TCP may support reliable configuration.
3. **IPv6 and ICMPv6:** IPv6 provides large addressing, while ICMPv6 supports control and diagnostic messages. Example: A smart meter receives an IPv6 address and uses ICMPv6 for network control.
4. **6LoWPAN-ND:** Neighbor discovery is optimized for constrained low-power networks. Example: A node discovers a nearby router without heavy conventional IPv6 overhead.
5. **RPL routing:** RPL routes IPv6 packets in low-power and lossy networks. Example: Meter data travels through multiple low-power routers to a border router.
6. **6LoWPAN adaptation layer:** It compresses and adapts IPv6 packets so they fit IEEE 802.15.4 frames. Example: A 40-byte IPv6 header is compressed before transmission.
7. **2.4 GHz band:** It provides 16 channels and a data rate of 250 kbps. Example: A home automation system uses 2.4 GHz worldwide.
8. **915 MHz band:** It provides 10 channels and a data rate of 40 kbps. Example: A regional low-power monitoring system may use 915 MHz.
9. **868 MHz band:** It provides 1 channel and a data rate of 20 kbps. Example: A European low-data-rate sensor network may use 868 MHz.
10. **Modulation methods:** The physical layer may use OQPSK, BPSK, or ASK depending on band and implementation. Example: OQPSK is commonly used at 2.4 GHz.

Others / Additional Relevant Points: The PHY frame includes synchronization information, PHY header, and a PHY service data unit of up to 127 bytes.

Conclusion: ZigBee IP creates an IPv6-capable low-power stack by combining application profiles, transport protocols, IPv6, RPL, 6LoWPAN, and IEEE 802.15.4.

Very Important Exam Suggestion: Draw the ZigBee IP stack from application profile down to IEEE 802.15.4 PHY. Write all three frequency/data-rate combinations exactly.

Question 3: Explain the IEEE 802.15.4 MAC layer, frame types, frame format, and security.

Answer (10 Marks)

Introduction: IEEE 802.15.4 defines the PHY and MAC foundation used by many low-power IoT technologies. The MAC layer controls how devices share the wireless medium and exchange frames.

1. **Channel access management:** The MAC layer coordinates access to the physical channel among nearby devices. Example: Several sensors avoid transmitting at the same moment.
2. **Network beaconing:** A coordinator sends beacons that help new devices identify and join the network. Example: A new temperature node detects the coordinator beacon.
3. **PAN association/disassociation:** The MAC layer manages joining and leaving a Personal Area Network. Example: A sensor associates with the building PAN during installation.
4. **Reliable peer communication:** Acknowledgements and frame checks improve link reliability. Example: A receiver sends an ACK after receiving a data frame.
5. **Data frame:** It carries application or network data. Example: A humidity reading is sent in a data frame.
6. **Beacon frame:** It advertises and synchronizes the PAN. Example: The coordinator periodically transmits a beacon.
7. **Acknowledgement frame:** It confirms successful reception. Example: The sender retransmits if no acknowledgement arrives.

8. **MAC command frame:** It carries control information between devices. Example: Association requests and responses use command frames.
9. **MAC frame fields:** The header contains Frame Control, Sequence Number, and Addressing Fields; the payload carries data; the footer contains the Frame Check Sequence. Example: The sequence number helps identify a particular frame.
10. **Security support:** When security is enabled, an Auxiliary Security Header is inserted and AES is used for confidentiality and integrity. Example: A smart lock encrypts its control messages.

Others / Additional Relevant Points: Addressing fields may contain destination/source PAN identifiers and destination/source addresses; addressing length varies with the selected mode.

Conclusion: The IEEE 802.15.4 MAC layer provides medium access, network participation, reliable frame exchange, multiple frame types, and link-layer security.

Very Important Exam Suggestion: Draw the MAC frame and label Frame Control, Sequence Number, Addressing Fields, Payload, FCS, and Auxiliary Security Header.

Question 4: Discuss IEEE 802.15.4e, IEEE 802.15.4g, and Wi-SUN for IoT.

Answer (10 Marks)

Introduction: IEEE 802.15.4e and 802.15.4g enhance low-power wireless networking for industrial automation, process control, smart grid, and smart utility applications.

1. **Need for enhancement:** Basic 802.15.4 has limitations such as MAC reliability issues, unbounded latency, and multipath fading. Example: A factory control packet may arrive too late in a noisy environment.
2. **IEEE 802.15.4e focus:** 802.15.4e expands and improves the MAC layer. Example: Industrial sensors obtain more predictable and reliable communication.
3. **MAC reliability:** The amendment improves dependable frame delivery. Example: A machine alarm is less likely to be lost.
4. **Latency improvement:** It addresses unpredictable or unbounded transmission delay. Example: A process-control actuator receives commands on time.
5. **Multipath fading handling:** Enhancements improve performance where reflected signals create fading. Example: A metal-filled factory has many reflected radio paths.
6. **Application domains:** It is designed for factory automation, process automation, and smart grid. Example: A smart-grid field device communicates reliably with a utility gateway.
7. **IEEE 802.15.4g focus:** 802.15.4g extends the physical layer for smart utility networks and larger frame support. Example: A utility field-area network carries meter data over a broader area.
8. **Larger PSDU:** The 802.15.4g PHY diagram permits a PSDU up to 2047 bytes. Example: A utility message can carry more information than a basic 127-byte frame.
9. **Wi-SUN meaning:** Wi-SUN stands for Smart Utility Network and is an alliance/ecosystem based on suitable IEEE standards. Example: Smart meters from different vendors operate in one utility network.
10. **Interoperability benefit:** Standardization supports multi-vendor deployment and scalable utility systems. Example: A city utility can replace one vendor device without redesigning the entire network.

Others / Additional Relevant Points: 802.15.4e mainly improves MAC behavior, while 802.15.4g mainly extends PHY capabilities for utility networks.

Conclusion: Together, 802.15.4e, 802.15.4g, and Wi-SUN provide a stronger foundation for reliable industrial and smart-utility IoT.

Very Important Exam Suggestion: Write a clear comparison: 802.15.4e = MAC enhancement; 802.15.4g = PHY enhancement; Wi-SUN = smart utility alliance/ecosystem.

Question 5: Explain IEEE 1901.2a Narrowband Power Line Communication and its applications.

Answer (10 Marks)

Introduction: IEEE 1901.2a-2013 is a wired IoT communication standard for Narrowband Power Line Communication (NB-PLC). It sends data over the same wires that carry electrical power.

1. **Wired constrained technology:** Unlike most constrained IoT access technologies, NB-PLC uses power lines instead of radio waves. Example: A smart meter sends data through existing electrical wiring.
2. **Narrowband spectrum:** It uses a narrow frequency band to support efficient communication. Example: Small utility messages are transmitted without wide bandwidth.
3. **Low-power operation:** The technology is suitable for devices with limited power requirements. Example: A utility meter communicates periodically with low energy use.
4. **Long-range capability:** Power-line infrastructure can carry communication over useful utility distances. Example: Meters in different parts of a neighborhood report to the utility system.
5. **Interference resistance:** NB-PLC is designed to work reliably despite interference on power lines. Example: Meter readings continue even when electrical appliances create noise.
6. **Smart metering:** It automates electricity, gas, or water meter reading. Example: A utility reads monthly consumption remotely.
7. **Distribution automation:** It monitors and controls grid devices. Example: A utility remotely checks and switches distribution equipment.
8. **Public lighting:** It manages city and street lighting. Example: Street lights report faults and receive switching commands.
9. **EV charging and microgrids:** It supports charging stations and local grids that may operate independently. Example: A charging station exchanges status data with the energy network.
10. **Renewable energy:** It can be applied to solar, wind, hydroelectric, and geothermal systems. Example: A solar inverter reports output through the power line.

Others / Additional Relevant Points: NB-PLC is especially attractive where power wiring already exists and radio communication is difficult or unnecessary.

Conclusion: IEEE 1901.2a provides low-power, long-range, and interference-resistant IoT communication over electrical power lines.

Very Important Exam Suggestion: Start by writing “wired technology” and “same wires carry power and data,” then list at least five applications.

Question 6: Describe IEEE 802.11ah (Wi-Fi HaLow), including PHY, MAC, topology, and applications.

Answer (10 Marks)

Introduction: IEEE 802.11ah, commonly called Wi-Fi HaLow, is a sub-GHz version of Wi-Fi designed to improve range, penetration, power efficiency, and large-device support for IoT.

1. **Reason for development:** Traditional Wi-Fi lacks sub-GHz penetration, very low-power operation, and efficient support for huge device populations. Example: Thousands of battery sensors are unsuitable for ordinary Wi-Fi behavior.
2. **Sub-GHz physical layer:** It operates in unlicensed sub-GHz bands that vary by region. Example: North America and parts of Asia-Pacific use 902-928 MHz ranges.
3. **Better range and penetration:** Lower frequencies travel farther and penetrate obstacles better. Example: A basement meter connects to an outdoor access point.
4. **Large device support:** One access point can support up to 8192 devices. Example: A large smart-building network connects thousands of sensors.
5. **Shortened MAC header:** Reduced overhead makes communication more efficient. Example: Small sensor packets carry less control overhead.
6. **Null Data Packet support:** Control information can be placed efficiently in PHY-related signaling, reducing decoding and energy cost. Example: A device handles control exchange without processing a full payload.
7. **Grouping and sectorization:** The AP groups stations and can use sectors to reduce contention. Example: Only one group in one sector competes during a selected window.
8. **Restricted Access Window:** RAW schedules groups so many devices do not transmit simultaneously. Example: Meter group A sends first, then group B.
9. **Target Wake Time:** TWT lets devices sleep until their assigned access time. Example: A sensor wakes every 30 minutes to report data.
10. **Topology and use cases:** It normally uses star topology and may use simple client relays; uses include smart-grid sensors, industrial aggregation, and extended-range Wi-Fi. Example: A client relays a distant agricultural sensor toward the AP.

Others / Additional Relevant Points: No special new security scheme is identified beyond the security mechanisms used in the wider IEEE 802.11 family.

Conclusion: Wi-Fi HaLow adapts Wi-Fi to IoT through sub-GHz range, power-saving MAC features, large device capacity, and controlled channel access.

Very Important Exam Suggestion: Underline 8192 devices, RAW, TWT, shortened MAC header, sub-GHz, star topology, and client relay.

Question 7: Define 6LoWPAN and explain its major features and addressing.

Answer (10 Marks)

Introduction: 6LoWPAN means IPv6 over Low-Power Wireless Personal Area Networks. It enables small, low-power, limited-processing devices to communicate using Internet Protocol over IEEE 802.15.4.

1. **IPv6 over low-power WPAN:** 6LoWPAN carries IPv6 communication on constrained wireless networks. Example: A battery sensor receives an IPv6-compatible identity.
2. **Internet connectivity:** It allows low-power devices to connect to the Internet. Example: A soil sensor sends data to a cloud server through a border router.
3. **IETF origin:** 6LoWPAN was created through IETF standardization work for constrained networking. Example: Implementations follow IETF RFC-based specifications.
4. **128-bit IPv6 support:** It enables IEEE 802.15.4 radios to carry IPv6 addressing. Example: Each IoT node can participate in a very large IP address space.

5. **Header compression:** Large IPv6 headers are compressed to reduce overhead. Example: A small sensor packet does not waste most of the frame on headers.
6. **Packet reformatting:** IPv6 packets are adapted to fit the IEEE 802.15.4 packet format. Example: An oversized IPv6 datagram is fragmented into smaller frames.
7. **Address translation/adaptation:** Adaptation techniques connect link-layer addressing with IPv6 operation. Example: A short PAN address is used efficiently inside the local network.
8. **64-bit extended address:** A 64-bit extended address is globally unique. Example: A device retains a unique hardware-level identity.
9. **16-bit short address:** A 16-bit short address is PAN-specific and assigned by the PAN coordinator. Example: A coordinator assigns address 0x1234 to a sensor.
10. **Application areas:** 6LoWPAN is used in IoT, smart grid, and machine-to-machine systems. Example: Smart meters communicate through an IPv6-enabled low-power mesh.

Others / Additional Relevant Points: Because IEEE 802.15.4 does not directly support IPv6 multicast, IPv6 packets may be carried as link-layer broadcast frames.

Conclusion: 6LoWPAN makes IPv6 practical for constrained IEEE 802.15.4 devices through compression, adaptation, addressing, and packet reformatting.

Very Important Exam Suggestion: Write the full form first, then draw IPv6 -> 6LoWPAN Adaptation -> IEEE 802.15.4. Mention both 64-bit and 16-bit addresses.

Question 8: Explain the need for the 6LoWPAN adaptation layer and packet handling.

Answer (10 Marks)

Introduction: The 6LoWPAN adaptation layer sits between IPv6 and IEEE 802.15.4. It solves the mismatch between large IPv6 packets/headers and the small 802.15.4 frame size.

1. **Size mismatch:** IPv6 packets can be much larger than the IEEE 802.15.4 maximum frame size. Example: A large IPv6 datagram cannot be placed directly in one 127-byte frame.
2. **Header compression:** The adaptation layer compresses IPv6 and related headers. Example: Predictable address parts are omitted and reconstructed later.
3. **Fragmentation:** A large packet is divided into multiple smaller IEEE 802.15.4 frames. Example: One IPv6 datagram becomes several radio fragments.
4. **Reassembly:** The receiver combines fragments to reconstruct the original packet. Example: A border router reassembles all numbered fragments.
5. **Dispatch identification:** Dispatch information indicates which type of 6LoWPAN header or payload follows. Example: The receiver recognizes whether the frame carries compressed IPv6.
6. **Mesh delivery support:** Optional mesh information can assist forwarding inside the low-power PAN. Example: A packet crosses multiple 802.15.4 nodes before reaching the border router.
7. **Efficient addressing:** It maps IPv6 operation to 64-bit extended or 16-bit short link addresses. Example: A short local address reduces packet overhead.
8. **Low processing overhead:** Compression and compact formats reduce transmission and processing burden. Example: A constrained sensor spends less energy sending headers.
9. **Interoperability with IP:** The original IPv6 packet is recovered at the network boundary. Example: Internet servers receive a normal IPv6 packet.
10. **Energy efficiency:** Fewer transmitted bytes reduce radio-on time, collisions, and battery consumption. Example: A periodic meter lasts longer on battery power.

Others / Additional Relevant Points: The adaptation layer is essential because direct use of full IPv6 over small IEEE 802.15.4 frames would be inefficient and often impossible.

Conclusion: Through compression, fragmentation, reassembly, and compact addressing, the 6LoWPAN adaptation layer makes IPv6 suitable for low-power networks.

Very Important Exam Suggestion: Draw a simple flow: IPv6 packet -> compression/fragmentation -> 802.15.4 frames -> reassembly -> IPv6 packet.

Question 9: Discuss routing considerations in 6LoWPAN.

Answer (10 Marks)

Introduction: Routing in 6LoWPAN must work in low-power and lossy networks where devices have limited memory, processing ability, battery power, and unreliable wireless links.

1. **Routing inside PAN:** The network must support multi-hop mesh routing among devices within the low-power PAN. Example: A distant sensor reaches the border router through two relay nodes.
2. **Routing to IPv6 domain:** Routes are also required between the PAN and external IPv6 networks. Example: Sensor data travels from the local mesh to an Internet server.
3. **Constrained nodes:** Routing state and control messages must be small. Example: A tiny sensor cannot store a large routing table.
4. **Constrained network:** Low bandwidth and lossy links require limited control overhead. Example: Frequent routing broadcasts would waste capacity.
5. **Energy awareness:** Routing should reduce transmissions and avoid unnecessary wake time. Example: A stable path is preferred over repeated route discovery.
6. **Link unreliability:** Routes must adapt when nodes or wireless links fail. Example: A new parent is selected after a relay goes offline.
7. **Scalability:** The protocol should support many devices without excessive signaling. Example: A smart-grid deployment contains thousands of meters.
8. **LOADng option:** LOADng provides reactive route discovery derived from AODV. Example: A route is discovered only when a source has data to send.
9. **RPL option:** RPL maintains a routing topology for low-power and lossy IPv6 networks. Example: Meters maintain upward routes toward a border router.
10. **Protocol selection:** Selection depends on traffic pattern, network stability, memory, latency, and energy needs. Example: Reactive routing may suit occasional traffic, while proactive topology maintenance may suit regular reporting.

Others / Additional Relevant Points: Bidirectional-link verification and failure recovery are important because LLN links may be asymmetric.

Conclusion: 6LoWPAN routing must balance reachability, low overhead, energy efficiency, reliability, and communication with the wider IPv6 Internet.

Very Important Exam Suggestion: Always mention the two routing areas: within the PAN and between the PAN and IPv6 domain. Then name LOADng and RPL.

Question 10: Explain the operation and important features of LOADng routing.

Answer (10 Marks)

Introduction: LOADng is the Lightweight On-Demand Ad Hoc Distance-Vector Routing Protocol - Next Generation. It is derived from AODV and extended for IoT and low-power networks.

1. **On-demand behavior:** Routes are created only when required. Example: A sensor starts route discovery only before sending data.
2. **RREQ generation:** The originator generates a Route Request for the required destination. Example: Node A broadcasts an RREQ to find node D.
3. **RREQ forwarding:** Intermediate LOADng routers forward the request until it reaches the destination. Example: Routers B and C relay the RREQ.
4. **Destination-only reply:** Only the destination is allowed to respond to an RREQ. Example: Even if router C knows a route, it cannot send the reply.
5. **RREP generation:** The destination creates a Route Reply after receiving the request. Example: Node D generates an RREP for node A.
6. **Hop-by-hop unicast reply:** The RREP returns through intermediate routers toward the originator. Example: D -> C -> B -> A.
7. **Route installation:** Nodes learn forwarding information as the request and reply travel. Example: Router B records the next hop for D.
8. **Route Error:** A broken route causes a Route Error message to be sent to the originator. Example: If link C-D fails, A receives an RERR.
9. **Optimized flooding:** RREQ flooding is optimized to reduce signaling overhead. Example: Duplicate or unnecessary requests are controlled.
10. **Sequence numbers:** RREQ and RREP messages from a router use a shared monotonically increasing sequence number. Example: Newer route information can be distinguished from older information.

Others / Additional Relevant Points: Intermediate routers are explicitly prohibited from replying to RREQs, which simplifies behavior but may increase route discovery distance.

Conclusion: LOADng provides lightweight, reactive route discovery and error reporting suitable for constrained IoT networks.

Very Important Exam Suggestion: Draw the exact message flow: Originator --RREQ--> Destination --RREP--> Originator; add RERR for route breakage.

Question 11: Explain RPL routing for low-power and lossy networks.

Answer (10 Marks)

Introduction: RPL is an IPv6 distance-vector routing protocol designed for Low-Power and Lossy Networks (LLNs), where devices and links are constrained and may fail frequently.

1. **IPv6 routing protocol:** RPL directly supports IPv6 routing in constrained networks. Example: An IPv6 smart meter routes data to a border router.
2. **Distance-vector approach:** Nodes make routing decisions using information received from neighboring nodes. Example: A sensor selects a suitable parent based on advertised routing information.
3. **Low-rate beaconing:** RPL maintains the routing topology using low-rate control beacons. Example: Stable nodes send control messages infrequently.
4. **Adaptive beaconing:** The beacon rate increases when inconsistencies are detected. Example: A failed link causes faster control messaging.

5. **Proactive function:** RPL proactively maintains a usable routing topology. Example: A route is generally available before periodic sensor data is generated.
6. **Reactive repair:** It reacts to inconsistencies to repair the topology. Example: A node selects another parent after its current parent fails.
7. **Routing information in datagrams:** Relevant routing information may be included with packets. Example: A packet carries information needed for forwarding behavior.
8. **Bidirectional-link requirement:** RPL operations require bidirectional links. Example: A child must be able to send to and receive from its parent.
9. **Asymmetric-link challenge:** LLN links may have different quality in opposite directions. Example: A node hears a router strongly, but the router receives the node weakly.
10. **Parent reachability verification:** A router must be verified as reachable before it is selected as a parent. Example: A node tests communication before using a neighbor for routing.

Others / Additional Relevant Points: RPL is appropriate for many regular data-collection networks, while its control behavior must still be tuned to avoid unnecessary overhead.

Conclusion: RPL combines proactive topology maintenance with reactive inconsistency resolution for reliable IPv6 routing in LLNs.

Very Important Exam Suggestion: Use the exact keywords: distance-vector IPv6, LLN, low-rate beaconing, adaptive beacon rate, proactive, reactive, bidirectional link, and parent verification.

Question 12: Compare ZigBee, Wi-Fi HaLow, NB-PLC, 6LoWPAN, LOADng, and RPL for IoT use.

Answer (10 Marks)

Introduction: Different IoT technologies solve different communication problems. A correct design selects the technology according to medium, range, power, addressing, topology, and routing requirements.

1. **ZigBee:** A low-power IEEE 802.15.4-based stack with application support, security, and star/tree/mesh networking. Example: Home lighting and sensor automation.
2. **Wi-Fi HaLow:** A sub-GHz Wi-Fi technology with long range, large device support, RAW, and TWT. Example: A large agricultural or smart-building sensor network.
3. **NB-PLC:** A wired narrowband technology that communicates over electrical power lines. Example: Smart metering and public lighting.
4. **6LoWPAN:** An adaptation technology that carries compressed IPv6 over IEEE 802.15.4. Example: Internet-connected low-power sensor mesh.
5. **LOADng:** A reactive, on-demand routing protocol derived from AODV. Example: Occasional traffic where routes need not be maintained continuously.
6. **RPL:** An IPv6 LLN routing protocol with proactive topology maintenance and reactive repair. Example: Regular smart-grid data collection.
7. **Communication medium:** ZigBee, Wi-Fi HaLow, and 6LoWPAN commonly use wireless links, while NB-PLC uses wires. Example: A radio-restricted utility room may favor NB-PLC.
8. **Power and scale:** ZigBee and 6LoWPAN suit highly constrained devices; HaLow supports many stations with scheduled wake times. Example: Thousands of sensors may use HaLow TWT.
9. **Internet integration:** 6LoWPAN provides direct IPv6 adaptation, while ZigBee may use its own network layer unless ZigBee IP is used. Example: An IPv6 cloud application benefits from 6LoWPAN or ZigBee IP.

10. **Final selection:** Choose according to the actual environment, traffic pattern, existing infrastructure, and reliability needs. Example: Use NB-PLC for power-line meters, ZigBee for home mesh, and RPL for IPv6 LLN routing.

Others / Additional Relevant Points: No single technology is best for every IoT system; hybrid systems may combine an access technology, adaptation layer, and routing protocol.

Conclusion: ZigBee, HaLow, NB-PLC, 6LoWPAN, LOADng, and RPL are complementary technologies selected according to specific IoT constraints.

Very Important Exam Suggestion: Make a comparison table with columns: Technology, Medium/Layer, Key Feature, Best Application. This earns presentation marks.

Final Rapid Revision Table

Topic	Core Keyword	Main Strength	Typical Example
ZigBee	AODV, APS, ZDO	Low-power mesh	Home automation
802.15.4	PHY + MAC	Low-rate WPAN	Sensor link
802.15.4e	MAC enhancement	Reliability/latency	Industrial automation
802.15.4g	PHY enhancement	Smart utility range	Utility network
1901.2a	NB-PLC	Power-line communication	Smart meter
802.11ah	HaLow, RAW, TWT	Sub-GHz, 8192 devices	Large sensor network
6LoWPAN	IPv6 adaptation	Compression/ fragmentation	IPv6 sensor mesh
LOADng	RREQ/RREP/RERR	Reactive routing	Occasional traffic
RPL	LLN, beaconing	Proactive + repair	Smart grid

Source Note: Prepared from the uploaded W9_compressed lecture slides and the CSE 441 Summer 2026 course outline. The answer structure is expanded for exam writing while preserving the source terminology and facts.